Electromagnetic Warfare (EW) BOLC

Comprehensive Exam Study Guide · synthesized from all 22 course documents

UNCLASSIFIED · guide · 4 practice tests · linked sources

Course: 17A/17-series Cyber & Electromagnetic Warfare Basic Officer Leader Course (BOLC) **Scope:** All 22 source documents — EW history, EMS fundamentals, EW math (logarithms & decibels), EM wave theory, radio-wave propagation, radio fundamentals, antenna theory, software-defined radios, EA/EP/ES doctrine, CEMA doctrine, threat assessment/IPOE, EW platoon force structure, movement, site selection, hide sites, reconnaissance, training management, contested-space operations, tactical radios, and historical case studies. **Classification of source material:** UNCLASSIFIED (one tactical-radio document is CUI). This guide is UNCLASSIFIED.

How to Use This Guide

Sections are grouped into six study blocks: **(1) History & Doctrine**, **(2) EMS & Wave Fundamentals**, **(3) EW Math**, **(4) Radio, Antenna & SDR Systems**, **(5) EW Divisions (EA/EP/ES)**, and **(6) Tactical Employment of the EW Platoon**. Each section lists must-know facts, key terms, and figures. A **Quick-Reference Tables** block and a **Self-Test Question Bank** with answers follow at the end. Bold items and tables are the highest-yield material for the exam.

BLOCK 1 — EW HISTORY & DOCTRINE

1. Evolution of Electronic/Electromagnetic Warfare (112-BOLCEW01)

Source slide deck (PDF): 112-BOLCEW01_-_EW_History_May_2024-4.pdf

1.1 Pre-World War I (Foundational Events)

- **1888 — Heinrich Hertz** demonstrated that electric sparks propagate through space (seen as a scientific phenomenon, not communications).
- **1897 — Guglielmo Marconi** invented the first radio; sent the first wireless communication over open sea (Bristol Channel → Flat Holm Island, ~3.7 miles). Patent approved, production began.
- **1901 — First deliberate radio jamming:** America's Cup Yacht Race (Columbia vs. Shamrock II). Competing wireless telegraph companies hired by newspapers; one held a transmitter down over an hour to jam competitors.

- **1903 — Marconi/Maskelyne Affair:** First **Electronic Attack**. Nevil Maskelyne ("Scientific Hooligan") injected a mocking Morse message into Marconi's "secure" Royal Institution demonstration.
- **1903 — Navy fleet exercise (Blue vs. White):** USS Texas jamming attempt; officer delayed the jam to hear the whole message — one of the first recorded **Intelligence Gain/Loss (IGL)** conflicts.
- **1904 — Russo-Japanese War:** First documented use of EW in **combat**. Russian operators used "Spark" transmitters to jam Japanese transmissions. Proof that **any transmitter is a jammer.** Russia commemorates 15 April as "Day of Radio Electronic Warfare."
- **1906 —** Army Signal Corps' first wireless platoon (Cuba to Key West).

1.2 World War I (1914–1918)

- British Cable Ship (CS) **Alert** cut German undersea telegraph cables → forced Germany onto radio → birth of **Information Warfare**.
- Birth of modern **SIGINT** — listening stations (e.g., Eiffel Tower); encryption became necessary. Army's Cipher Bureau **MI-8** established by **Herbert Yardley**.
- **1915 —** UK Royal Navy develops **Direction Finding (DF)**.
- **Clapper-break** on aircraft = first use of **Electronic Counter-Countermeasures (ECCM/EP)**.

1.3 Interwar & World War II

- **RADAR = RAdio Detection And Ranging.** 1935 — Robert Watson-Watt bounced a radio wave off an aircraft (~17 mi); 1936 range improved to 75 mi. UK "Chain Home" early-warning network.
- **1938 —** first airborne jamming of radar (Germans).
- **1941 —** US established the **Radio Research Laboratory (RRL)** (originally the Radar Counter-Measure Laboratory; renamed for secrecy). RRL developed virtually every US jammer/receiver/countermeasure of WWII.
- **Battle of the Beams (1940–41):** Germany's "Knickebein" navigation beams for night bombing; Britain countered via airborne ELINT + jamming.
- **Ferret Flights (1943):** bombers turned into ELINT aircraft to detect enemy radar.
- **Chaff / "Window" / "Rope":** foil strips creating deceptive radar echoes. D-Day (**Operations TAXABLE and GLIMMER**, 5–6 Jun 1944) simulated a fake fleet toward Pas-de-Calais and blinded German radar during the Normandy landings.
- Army's limited ground jamming stemmed from **Intelligence Gain/Loss** — jamming would interfere with vital communications intelligence.

1.4 Cold War, Korea, Vietnam

- **1945 —** Signal Security Agency became the **Army Security Agency (ASA)** under the Director of Military Intelligence.
- **1954 —** 72nd and 73rd Communications EW Battalions formed at Fort Huachuca, AZ. 1955 Operation SAGEBRUSH — 73rd's jamming so effective they were asked to leave.
- **1955 —** ASA assumed control of ELINT and communications ECM from the Signal Corps.
- **Vietnam (1961–1975):** ASA entered May 1961. **Spec-4 James Davis** — sometimes called "First U.S. Serviceman Killed in Combat in Vietnam" (DF mission). **SEAD = Suppression of Enemy Air Defense** matured; F-100/F-105 "**Wild Weasels**" used against SA-2s and radar.

- **1982 — First Lebanon War / Operation Mole Cricket 19 (Bekaa Valley):** first time a Western-equipped air force destroyed a Soviet-built SAM network (Israel vs. Syria), using ELINT and RPVs.

1.5 1990s–Present

- **Desert Shield/Storm (1990–91):** "**Spoof and Punch**" — ELINT mapped Iraqi IADS; drones/decoys/jammers suppressed; stealth destroyed C2 nodes. Air victory in 96 hrs; only 2 aircraft lost of ~900 flying 38,000 sorties. Ground EW (**CEWI companies, TLQ-17A jammer**) was underwhelming.
- **Post-9/11:** OEF (Oct 2001), OIF (Mar 2003). **RCIEDs** became the single most effective weapon vs. US forces (>7,700 casualties by May 2005). Response = **CREW (Counter-RCIED Electronic Warfare)** under new **Functional Area (FA) 29**.
- **FA 29 career field (2009):** 29-series → FA 29 EW Officer, 290A EW Technician (WO), 29E EW Specialist (NCO).
- **EW → Cyber Branch:** Jan 2014 transition announced; 2016 EW School became "EW College" under Cyber School; **1 October 2018 — EW officially became part of Cyber Branch.** New MOS: **17B (Officer), 170B (Warrant/Technician), 17E (Specialist/NCO).** Training consolidated at Fort Gordon (2020/2021).
- **EW Insignia** approved 26 October 2011. **Association of Old Crows (AOC):** WWII ECM officers were "Ravens" → instructors became "Old Crows"; AOC founded 1964; journal became *Journal of Electronic Defense (JED)* in 1978.
- **Modern conflicts:** 2014 Russia/Ukraine (CEMA + disinformation); **2020 2nd Nagorno-Karabakh War** (drones, loitering munitions, GNSS jamming); **2022–present Russo-Ukraine War** (small-UAS warfare, GNSS denial, EMS discipline).
- **Emerging tech:** Directed-energy weapons, AI for spectrum management, **C-UAS**, JADC2, Mosaic Warfare, **CHAMP** (high-power microwave), **LaWS** (laser), EMP rifles.

1.6 Overarching Historical Lesson (from "Case Studies" monograph — MAJ Henke)

Source document (PDF): Case Studies.pdf

The US Army has repeatedly **neglected EW between conflicts and rebuilt it under crisis.** Three lessons:

1. Offensive/defensive **EA and ES work** when resourced, trained, and integrated (Overlord, Vietnam ARDF, Desert Storm aviation).
2. **EW capabilities atrophy** between conflicts if not maintained.
3. Failing to maintain EW risks **unnecessary loss of personnel and equipment** at the start of the next conflict — critical given the speed/lethality of future LSCO vs. Russia/China.

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2. Army CEMA & EW Doctrine (NEW Army CEMA Doctrine — 112-BOLCEW03)

Source slide deck (PDF): NEW-_Army_CEMA_Doctrine-2.pdf

2.1 Core Definitions

- **CEMA (Cyberspace Electromagnetic Activities):** actions leveraged to **seize, retain, and exploit** an advantage over adversaries in **both cyberspace and the EMS**, while denying/degrading enemy use of the same and protecting the mission command system.

- **CEMA is comprised of:** Electromagnetic Warfare + Cyberspace Operations + Spectrum Management Operations.
- **Electromagnetic Warfare (EW):** military action involving the use of electromagnetic and directed energy to **control the EMS or to attack the enemy** (FM 3-12). **Purpose:** deny the enemy control of the EMS while ensuring friendly freedom of maneuver within it.
- **EW's three divisions: Electromagnetic Attack (EA)**, **Electromagnetic Protection (EP)**, **Electromagnetic Support (ES).**

2.2 Related Concepts

- **Spectrum Management Operations (SMO):** spectrum management + frequency assignment + host-nation coordination + policy. SMO is the management portion of EMSO.
- **EMSO (Electromagnetic Spectrum Operations)** = EW + SMO combined.
- **Warfighting Functions (WfF):** Movement & Maneuver, Intelligence, Fires, Sustainment, Protection, **Mission Command (C2)**. Integration/synchronization of CEMA is a task of the **Mission Command** WfF.

2.3 Key CEMA/EW Publications

Publication	Title / Purpose
FM 3-12	Cyberspace and Electromagnetic Warfare Operations — TTPs for coordinating/integrating/synchronizing Army CEMA
ATP 3-12.3	Electromagnetic Warfare Techniques — how to conduct EW in operations
ATP 3-12.4	Electromagnetic Warfare Platoon — planning/preparing/executing EW PLT ops
JP 3-13.1	(Joint) Electronic/Electromagnetic Warfare
ADP/ADRP 6-0	Mission Command
ADP/ADRP 3-0	Operations
TC 3-12.1.2.2 (CEWTS)	US Army Cyber & EW Training Strategy

2.4 CEMA Force Structure (Corps and Below)

- **CEMA Cell** (Bde–Corps): CEMA OIC (17B), EWO (17B), CWO (17A), EW Technician (170B), Enlisted EW Staff (17E), Spectrum Manager (25E).
- **BDE EW Platoon mission:** conduct **electromagnetic sensing and effects delivery**; support the commander's **PIRs** and provide **early warning** to protect the force.
- Terms "CEMA" and "EW" are used interchangeably at Corps-and-below because almost all organic CEMA capability there is EW-related, and most CEMA personnel are EW (17E/170B/17B).

BLOCK 2 — EMS & ELECTROMAGNETIC WAVE FUNDAMENTALS

3. Introduction to the Electromagnetic Spectrum (112-BOLCEW02)

Source slide deck (PDF): 112-BOLCEW02_-_Introduction_to_EMS-2.pdf

3.1 Wave Characteristics

- **Frequency:** number of complete cycles/oscillations passing a fixed point per second. Unit = **Hertz (Hz)**. Waves are generated by an **alternating current**.
- **Amplitude:** height of a sine wave. *Peak* = reference line to peak/trough; *Peak-to-Peak* = max positive to max negative.
- **Cycle:** one complete phase (0 to 2π). **Peak/Crest** = highest point; **Trough** = lowest point.
- **Wavelength (λ):** distance between repeating points of a wave (usually meters). Symbol = lambda **λ**.
- **KEY RELATIONSHIP:** Frequency and wavelength are **inversely proportional**: **High Freq = Short λ; Low Freq = Long λ.**

3.2 The EMS (sorted by frequency, low → high)

Band	Frequency Range
Radio Waves (RF)	3 Hz – 300 GHz (≈0–300 MHz "radio waves" portion)
Microwaves	300 MHz – 300 GHz
Infrared (IR)	300 GHz – 405 THz
Visible Light	405 THz (Red) – 790 THz (Violet)
Ultraviolet (UV)	790 THz – 30 PHz
X-rays	30 PHz – 30 EHz
Gamma Rays	> 10 EHz

3.3 RF Sub-Bands ("Where We Operate" — memorize order)

From lowest to highest: **ELF, SLF, ULF, VLF, LF, MF, HF, VHF, UHF, SHF, EHF.** (UHF/SHF/EHF are microwaves.)

Band	Range	λ	Key Uses
ELF (Extremely Low)	3–30 Hz	10,000–100,000 km	Submarine comms (one-way only)
SLF (Super Low)	30–300 Hz	1,000–10,000 km	AC power grids (50/60 Hz); submarine comms
ULF (Ultra Low)	300 Hz–3 kHz	100–1,000 km	Mine comms; penetrates earth
VLF (Very Low)	3–30 kHz	10–100 km	Submarines (penetrates water 10–40 m); nav beacons
LF (Low)	30–300 kHz	1–10 km	AM broadcast (Europe/Africa/Asia); aircraft beacon/nav
MF (Medium)	300 kHz–3 MHz	100–1,000 m	Maritime distress (500 kHz); AM radio
HF (High)	3–30 MHz	10–100 m	Medium/long-distance comms; skywave
VHF (Very High)	30–300 MHz	1–10 m	FM radio/TV, aircraft; ideal short-distance **LOS**; tropospheric ducting
UHF (Ultra High)	300 MHz–3 GHz	1 dm–1 m	TV, mobile phones, **GPS**, WLAN/WiFi
SHF (Super High)	3–30 GHz	1 cm–1 dm	Radar, SATCOM up/downlink, microwave links
EHF (Extremely High)	30–300 GHz	1 mm–1 cm	Prone to atmospheric attenuation; jam-resistant (beamforming/adaptive nulling)

Above radio/microwaves: **IR** (near IR in remotes; far IR = thermal/radiant heat), **Visible Light**, **UV** (dangerous, cellular-size λ), **X-rays** (penetrate skin), **Gamma Rays** (nuclear/radioactive; CMEs & space weather).

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4. Fundamentals of EM Waves (BOLCEW06)

Source slide deck (PDF): BOLCEW06-EM Wave Fundamentals.pdf

4.1 What is an EM Wave?

A **self-propagating energy wave** comprised of an **electric field and a magnetic field** that moves at the **speed of light.**

Fundamentals of an EM wave: - Moves at a fixed speed (speed of light). - Comprised of an electric AND magnetic field (perpendicular to each other and to direction of travel). - Has **no mass** (all energy, no matter). - Has a **net zero charge.** - **Self-propagating** — requires **no medium** (unlike normal/mechanical waves that need a medium such as water). - Acts as both a **particle (photon)** and a **wave**.

4.2 Wave Formulas (MUST MEMORIZE)

- **Speed of light (c) = 300,000,000 m/s = 3×10^8 m/s.**
- **$V = \lambda F$** (Velocity = wavelength × frequency)
- **$F = V/\lambda$** or **$F = c/\lambda$** (Frequency)
- **$\lambda = V/F$** or **$\lambda = c/F$** (Wavelength)
- **$F = 1/T$** (Frequency = inverse of the period, T)
- Worked example: EM wave with $\lambda = 1$ m → $F = (3\times10^8)/1$ = **300 MHz**. Conversion: 2000 kHz = 2 MHz.

4.3 Fields

- **Electric Field:** energy field around charged particles (measured in **volts**); field lines run **+ to −**.
- **Magnetic Field:** created by a **changing/moving electric field** (does not exist on its own); lines run **North to South**; cannot exist as a **monopole**.
- **Electromagnetic Force:** produced by changing/interacting/moving electric & magnetic fields; **perpendicular to both**.
- **Discovery timeline:** Gilbert (1500s magnetism) → Franklin (1752 charge) → Coulomb (1785 electric force) → Oersted (1820 electricity↔magnetism) → Faraday (1830 induction) → **Maxwell (1861 unified E&M).**

4.4 Polarization

- Determined by the **orientation of the electric field.**
- **Vertically polarized:** E-field lines at right angles to Earth's surface (need a vertical antenna).
- **Horizontally polarized:** E-field lines parallel to Earth's surface.
- **Circular polarization** combines both.
- **Right-Hand Rule:** thumb = E field, forefinger = H (magnetic) field; your antenna is the electric component.

4.5 EM Wave Interactions (memorize the 4 core + Free Space Loss + Doppler)

- **Reflection:** bouncing off a surface (multipath, ground-reflected waves); phase shift occurs (~180°).
- **Refraction:** bending as a wave enters a medium of different density (wave must enter at <90°); enables beyond-LOS comms (tropo scatter, prism).
- **Diffraction:** bending **around** an object (terrain masking; part of the RF wave changes direction from LOS).
- **Absorption:** loss of energy as the wave passes through an object/medium (weather increases it; more significant at **higher frequencies**).
- **Free Space Loss:** signal-strength loss over a LOS path through free space, due to **wavefront spreading** (like a flashlight) — energy decreases as distance increases.
- **Doppler Effect (EM):** frequency change as Tx/Rx move relative to each other. Moving **toward** = higher freq = **blue shift**; moving **away** = lower freq = **red shift.** Used in radar, target detection, navigation, ES.

5. Radio Wave Propagation (BOLCEW07)

Source slide deck (PDF): BOLCEW07-Radio Wave Propogation.pdf

5.1 Free Space vs. On Earth

- **On Earth:** obstacles (trees/hills/cities), weather, charged atmospheric layers, curvature reduces LOS (~**6 miles** near the surface), man-made & natural interference.

5.2 Transmission Paths

- **Ground Waves** — Surface, Direct, and Ground-Reflected waves; affected by terrain, Earth's electrical characteristics, diffraction along curvature.
- *Direct wave:* LOS path; affected by curvature/terrain.
- *Surface wave:* travels along Earth's surface; attenuated by induced voltage & terrain conductivity; slightly longer than direct.
- *Ground-reflected wave:* reflects off Earth; can help go over obstacles; shorter coverage; may cause unwanted reception.
- **Sky Wave (Ionospheric Wave):** radiated upward, refracted back to Earth by the ionosphere — great distances.
- **Space Wave:** radio waves that leave Earth.

5.3 Atmosphere Layers

- **Troposphere:** where weather occurs; big effect on propagation.
- **Stratosphere:** between tropo and iono; constant temp; little/no effect; calm.
- **Ionosphere:** ionized by solar radiation; **4 charged layers during the day, 2 at night**; **refracts radio waves back to Earth.**

5.4 Wave Degradation / Transmission Losses

- Three degradation types: **Weather, Transmission Loss, Electromagnetic Interference (EMI).**
- Transmission losses: **Ground Reflection Loss** + **Free Space Loss** (+ Absorption). Combined absorption + ground-reflection + free-space loss account for most energy loss in ionospheric propagation. **Greatest loss of radio-wave energy is due to absorption.**
- **Weather:** rain attenuates by scattering (worse at higher freq) & absorbs; fog serious absorption **>2 GHz**; rain is **8× denser than snow** (snow attenuates less); hail depends on stone size.
- **Ducting (Temperature Inversion):** waves trapped between differing air temps — great distances, **Over-The-Horizon (OTH).**
- **Tropospheric Scattering:** enables VHF/UHF far beyond LOS; OTH reception up to **~500 miles.**

5.5 EMI

- **EMI:** interference producing annoying/impossible operating conditions; man-made (transmitters, motors, ignition, switches; worse near industrial areas, lower at night) or natural.

- **Controlling EMI:** cut antennas to correct frequency, limit bandwidth, use filtering/shielding; physically separate Tx/Rx antennas; use directional antennas.

BLOCK 3 — EW MATH

6. Logarithms (112-BOLCEW05)

Source slide deck (PDF): 112-BOLCEW05_-_Logarithms-3.pdf

6.1 Exponentiation Rules (base 10)

- $10^n = 10 \times 10 \times \ldots$ (n tens); $\mathbf{10^0 = 1}$; $10^{-n} = 1/10^n$.
- $10^m \times 10^n = 10^{(m+n)}$; $(10^m)^n = 10^{(m \cdot n)}$; $10^m/10^n = 10^{(m-n)}$.

6.2 Logarithms

- A **logarithm is the inverse of exponentiation.** If $b^L = x$, then **$\log_b x = L$** (b = base, L = exponent/logarithm, x = result).
- Examples: $\log_{10} 1000 = 3$; $\log_{10} 10 = 1$; $\log_{10} 400 \approx 2.60$.

6.3 Logarithmic Rules

1. **$\log(N \cdot M) = \log N + \log M$** (product → sum)
2. **$\log(N^M) = M(\log N)$** (power → multiply)
3. **$\log(N/M) = \log N - \log M$** (quotient → difference)

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7. Decibels (112-BOLCEW05) — HIGH-YIELD

Source slide deck (PDF): 112-BOLCEW05_-_Decibels-2.pdf

7.1 Power Ratio & Decibel Definition

- **Power ratio** = P_out / P_in (relative difference between two power levels). Example: 4 W / 2 W = 2 (P_out twice as strong).
- **Decibel (dB):** power ratio on a **logarithmic scale**: **$dB = 10 \log_{10}(P_{out} / P_{in})$.** dB values always followed by the "dB" symbol; no symbol = numerical form.

7.2 The "3s and 10s" Rules of Thumb (memorize)

- Multiplying the numerical power ratio **× 2** → **add 3 dB**; dividing **÷ 2** → **subtract 3 dB.**

- Multiplying **× 10 → add 10 dB**; dividing **÷ 10 → subtract 10 dB.**
- Reference values: ratio 2 ↔ 3 dB; 4 ↔ 6 dB; 8 ↔ 9 dB; 5 ↔ 7 dB; 10 ↔ 10 dB.
- **Convert dB → ratio:** ratio = 10^(N/10) (antilog). Ex: 46 dB ≈ 40,000; −13 dB ≈ 0.05.

7.3 Power Levels — dBW and dBm

- **dBW** = power referenced to **1 watt**: **P(dBW) = 10 log(P in watts).**
- **dBm** = power referenced to **1 milliwatt**: **P(dBm) = 10 log(P in mW).**
- Example: 500 mW = 10 log(500) ≈ **26.99 dBm.**
- **Conversion rule (MEMORIZE): N dBW = N dBm − 30 dB**, i.e., **dBW = (dBm − 30)**; dBm = dBW + 30.

7.4 Rules for Manipulation

- Adding/subtracting non-suffixed **dB** to/from dBm (or dBW) keeps the reference (result stays dBm/dBW). Ex: 10 dBm − 3 dB = 7 dBm; 24 dBW + 6 dB = 30 dBW.
- **dBm − dBm = dB**; **dBW − dBW = dB.** Ex: 15 dBm − 5 dBm = 10 dB.
- To subtract dBm from dBW (or vice versa) you must first convert references.
- **Do NOT add dBm or dBW to themselves or each other.**

7.5 Gain and Loss

- **Gain (g)** = P_out / P_in; in dB: **g(dB) = 10 log(P_out/P_in).** Ex: 1 W in → 100 W out → g = 100 → 20 dB. Gain of 350 ≈ 25.44 dB.
- **Loss (α)** = P_in / P_out = 1/g; in dB, **α(dB) = −g(dB)** (loss is the inverse of gain). Ex: 10 W → 1 W = 10 dB loss.
- **Systems with multiple gains/losses:** overall system gain = $(g_1 \cdot g_2 \cdot g_3 \ldots)/(\alpha_1 \cdot \alpha_2 \ldots)$. In dB, **add all gains, subtract all losses.** Example: g_1=500, g_2=100, g_3=10, α_1=20, α_2=5 → ratio 5000 → ≈ 36.99 dB.
- Output example: P_in = 10 dBW, g_system = 23 dB → **P_out = 33 dBW** (P_out dBW = P_in dBW + g_system dB).

BLOCK 4 — RADIO, ANTENNA & SDR SYSTEMS

8. Fundamentals of Radio Communications (112-BOLCEW08)

Source slide deck (PDF): 112-BOLCEW08-Radio Fundamentals (1).pdf

8.1 Layers & Components

- **Key layers:** Radio (Transmitter / Receiver / Transceiver) → Radio Link → Channel → Network.
- **Radio types:** Transmitter (Tx only), Receiver (Rx only), **Transceiver** (both).

- **Basic components:** power source, antenna, feed line, modulator/demodulator, wave generator (**oscillator**), **mixer**, user interface (mic).
- **Radio Link:** a wireless circuit formed by ≥2 radios (Tx, Tx lines, Tx antenna, propagation path, Rx antenna, Rx).
- **Channel:** bandwidth of frequency a link occupies. Key terms: **Bandwidth** (range a channel occupies), **Carrier Frequency** (center unmodulated freq), **Modulation** (how the carrier is changed), **Channel Spacing** (bandwidth between two channels).
- **Network:** a collection of radios linked by ≥1 channels.

8.2 Duplexing

- **Simplex (Broadcast):** one direction only (TV, commercial radio, mass messaging).
- **Half-Duplex (HDX):** both directions but **one at a time** (same frequency).
- **Full-Duplex (FDX):** both directions **simultaneously**; uses **two different frequencies** (phones, two-way radios).

8.3 Modulation & Mixing

- **Modulation:** changing a constant-frequency carrier wave according to a changing input wave by mixing the two signals.
- **Heterodyne (mixing):** combining a low-freq wave (audio/input) with a high-freq wave (carrier) to produce a higher-freq resultant. **Reginald Fessenden** demonstrated the first heterodyne receiver in 1901.
- **Carrier wave:** unmodulated wave modified to carry information; its center frequency is the radio's carrier.
- **Analog info:** continuous measurement. **Digital info:** sampled, quantized into discrete states (1,0).
- **Three major modulation types: AM** (Amplitude Modulation — incl. DSB, SSB, ASK), **FM** (Frequency Modulation — incl. FSK), **PM** (Phase Modulation — incl. PSK).

8.4 Selective Squelch

- **Squelch:** a circuit/noise gate that cuts off receiver output when no input (or below a threshold) is present.
- **Tone squelch / selective-calling techniques: CTCSS** (most common — superimposes ~50 sub-audible tones, 67–257 Hz), **SELCALL**, **DCS**, **XTCSS**, **DTMF** (Dual-Tone Multi-Frequency, 4×4 keypad matrix; used for selective calling and PTT unit ID).

8.5 COMSEC & Frequency Hopping

- **COMSEC:** encrypting radio comms so signals can only be demodulated with a shared key/cipher. All military radios should use it. No technique is 100% effective.
- **Frequency Hopping (Frequency Agile):** continuously moving through predesignated frequencies during transmission; requires shared protocol/precise timing (COMSEC). Reduces interference, hardens against jamming, lowers probability of intercept, harder to geolocate via DF. **Combination of frequency hopping + COMSEC = multiple layers of security.**

8.6 COTS Radio Services

- **FRS** (Family Radio Service): short-range, no license needed.

- **GMRS** (General Mobile Radio Service): land-mobile UHF short-distance.
- **PMR** (Professional Mobile Radio): single-organization networks (police/fire); VHF/UHF; ~2.0 W, up to 25 mi; 22 channels × 121 privacy codes = 2,622 combos; point-to-multipoint.

8.7 Common Military Radios (know band + echelon)

Radio	Type / Band	Primary Use	Lowest Echelon	Mod / Power
AN/PRC-150	HF, 3–30 MHz	Beyond-LOS voice/data	Company	AM, ASK / 20 W
AN/PRC-1523E	FM/ASIP, 30–88 MHz	LOS voice/data	Platoon	FM, FSK / 50 W
AN/PRC-117G	TACSAT, ~200–300 MHz	LOS voice/data	Company	FM, FSK / 10 W (UHF)
JBCP/BFT	~1600 MHz	Battle tracking/sync	Lt Co / Mounted Plt	FSK / 8 W
TRILOS	UHF band	LOS tactical internet	Battalion	FSK, PSK
STT (Satellite Transport Terminal)	SHF band	BLOS tactical internet transport	Battalion	PSK

8.8 Common SDRs (intro)

- **RTL-SDR:** 500 kHz–1.75 GHz; computer-based scanner (police/EMS/aircraft/GPS).
- **HackRF One:** 1 MHz–6 GHz; testing/monitoring; half-duplex.
- **KrakenSDR:** 24 MHz–1.766 GHz; direction finding & beamforming.

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9. Antenna Theory (112-BOLCEW09)

Source slide deck (PDF): 112-BOLCEW09-Antenna_Theory.pdf

9.1 Fundamentals

- **Antenna:** the interface between EM waves propagating through space and electric currents in system components. Tx supplies amplified current → antenna radiates EM energy; Rx intercepts EM waves → produces current.
- **Induction (key concept):** a current in a wire produces a magnetic field; a changing magnetic field creates a force that moves electrons (current).
- **Induction Field (near field):** localized; does **NOT** transmit EM energy; responsible for resonation effects reflected back to the generator.

- **Radiation Field (far field): responsible for radiation/transmission**; determines coverage pattern & directional strength; decreases as distance from antenna increases.

9.2 Key Electrical Concepts

- **Impedance (Z):** total opposition to AC in a conductor (resistance + magnetic-induction effects). **Z = E/I** (voltage/current). Critical to antenna design; incorrect impedance = low performance or damage.
- **Resonance:** an antenna's enhanced ability to efficiently emit/receive at certain frequencies. Occurs when the antenna is **½ the wavelength** of the target frequency (also at other multiples: ¼, ⅛, etc.). Smaller freq → larger antenna (HF); larger freq → smaller antenna (cell phone). **Resonant length L = C / (2f)** (C = 3×10^8 m/s, f = target freq in MHz).
- **Standing Wave Ratio (SWR):** ratio of outgoing energy vs. energy reflected back; high SWR damages/degrades performance. Related to impedance, resonance, and power output. **Match antenna to radio and frequency; don't exceed antenna power rating.**

9.3 Polarization (Antenna)

- Antennas are polarized based on the emitted electric field: **Linear** (vertical/horizontal), **Circular** (left/right hand), **Elliptical**.
- **Reciprocity:** antennas must match polarization for max coupling. Vertical vs. horizontal = **cross-polarization → 30 dB loss (catastrophic).** Circular waveform into a linear antenna = **3 dB loss.**

9.4 Radiation Pattern Terms

- **Main Lobe:** direction of maximum radiation. **Nulls:** directions of no radiation (0 dB). **First Null Beam Width (HPBW):** angle between first nulls. **Side Lobes:** lobes outside the main lobe. **Front/Back (F/B) Ratio:** main-beam signal vs. back signal.
- **Directionality/Gain:** bias to radiate in a specific direction; measured in **dBi (decibels isotropic)**; relative gain; used for DF and defeating interference.
- **Azimuth:** angle in the horizontal plane. **Elevation:** angle above horizontal.

9.5 Antenna Types (know directionality, polarization, gain, use)

Antenna	Directionality	Gain (typical)	Applications
Isotropic	Truly omni (theoretical, no nulls)	0 dBi	Benchmark for gain comparison
Half-Wave Dipole	Omni	~2 dBi	WLAN, VHF TV "rabbit ears", FM
Monopole (Whip)	Omni	~2 dBi	Military radios (most common), PTT
Log Periodic (LPA)	Directional (mod–high)	5–10 dBi	Spectrum analyzers, interference hunting, TV
Yagi	Directional (mod–high)	5–10 dBi	**Direction Finding**, extended LOS, SCADA
Conical Spiral	Directional (circular pol.)	8+ dBi	Point-to-point, microwave links
Parabolic Dish	Extremely directional	10+ dBi (up to 45 dBi large)	SATCOM, point-to-point backhaul
Phased Array (Fan Beam)	Electronically steerable	—	Military radar (beam steered via phase shifters)

- Gain levels: **High >20 dBi** (parabolic dish); **Medium 10–20 dBi** (horn, helix, Yagi); **Low <10 dBi** (dipole, loop, patch, slot, whip).

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10. Software Defined Radios (SDR) (112-BOLCEW11)

Source slide deck (PDF): 112-BOLCEW11_-_(U)_Software_Defined_Radios_(SDR).pdf

10.1 Definition & Concept

- **SDR:** a radio where components normally built into hardware (**mixers, filters, amplifiers, modulators/demodulators**) are implemented via **software** on a PC or embedded system. Advantages: **flexibility** and **ease of adaptation.**
- Ideal receiver: antenna → ADC → DSP → software application. Ideal transmitter: DSP generates numbers → DAC → antenna.
- **History:** "digital receiver" coined 1970 (DoD "Gold Room," tool called **Midas**); "software radio" coined 1984 (E-Systems, now Raytheon). **Project SpeakEasy** (Air Force, Phase 1, 1990–1995) aimed to operate 2 MHz–2 GHz; **Stephen Blust** (1995).

10.2 Key Principles

- **Nyquist Theorem:** to reproduce a signal, sample at a rate ≥ **2× the highest frequency (2f_max).** Undersampling → **aliasing** (false frequency components/distortion). Oversampling = much greater than 2×; wastes storage.
- **IQ Data:** complex samples with a real (**I, in-phase**) and imaginary (**Q, quadrature**) component; IQ sampling **doubles sampling bandwidth**; does not "break" Nyquist.
- **RF Mixing:** a frequency mixer creates new frequencies (sum and difference of two input signals).

10.3 Filtering & Resampling

- **Analog filters:** Low-pass (blocks above cutoff), High-pass (blocks below cutoff).
- **Digital filters:** FIR (Finite Impulse Response) & IIR (Infinite Impulse Response — fewer terms/calculations).
- **RF filter types (4):** Low-pass, High-pass, Band-pass, Band-reject.
- **Resampling: Decimation** (reduce sampling rate) and **Interpolation** (increase sampling rate).
- **Demodulation:** extracting the original info from the carrier. AM = envelope detection; FM = quadrature demodulation; PM = separating I & Q.

10.4 HackRF One (know specs)

- Frequency: **1 MHz – 6 GHz**; ADC resolution: **8 bits**; max bandwidth: **20 MHz**; **half-duplex** (TX and RX, not simultaneous); sample rates 2–20 Msps (quadrature); **open source**, portable.
- **JTRS** (Joint Tactical Radio System) — US military SDR program using an open **Software Communications Architecture (SCA).**

BLOCK 5 — THE THREE DIVISIONS OF EW (EA / EP / ES)

11. Electromagnetic Attack (EA) — (EA_ES_EP, 112-BOLCEW03)

Source slide deck (PDF): EA_ES_EP.pdf

11.1 Definition

- **EA:** use of EM energy, directed energy, or **anti-radiation weapons** to attack personnel, facilities, or equipment with intent to **degrade, neutralize, or destroy** enemy combat capability; considered a **form of fires** (JP 3-13.1).
- **D4M:** EA denies, degrades, disrupts, destroys, or **manipulates** enemy combat capability.

11.2 Offensive vs. Defensive EA

- **Offensive EA:** attacks an enemy capability to affect enemy personnel/equipment (jamming C2, anti-radiation missiles for SEAD, directed-energy).

- **Defensive EA:** attacks an enemy capability to **protect friendly** forces (flares/chaff/active decoys, towed decoys, DE-IR countermeasures, **Counter-RCIED**). *(Note: EP ≠ Defensive EA — see §12.)*

11.3 The 7 EA Actions/Missions

1. **Countermeasures** (active/passive; EO-IR & RF)
2. **Electromagnetic Deception** — 3 types: **Simulative** (represent friendly notional/actual capabilities), **Manipulative** (convey misleading indicators), **Imitative** (mimic threat emissions)
3. **Electromagnetic Intrusion** (insert EM energy into transmission paths to deceive/confuse operators)
4. **Electromagnetic Jamming**
5. **Electromagnetic Probing** (intentional radiation to learn enemy system functions/capabilities)
6. **Electromagnetic Pulse (EMP)** (strong pulse inducing damaging current/voltage surges)
7. *(Countermeasures + the above comprise the EA action set)*

11.4 Jamming — HIGH-YIELD

- **Fundamental principle:** raise the noise level at a receiver so it exceeds the incoming signal. **Jamming affects RECEIVERS, not transmitters.** A jammed unit can still transmit but may not receive.
- **Signal-to-Noise Ratio (S/N):** for a receiver to detect a signal, S/N must be **> 1**. If S/N < 1 → signal not detected (jammed).

Jamming techniques by employment:

Technique	Description
Standoff jamming (SOJ)	Supports ops by disrupting/degrading threat C2 and sensors from a distance
Escort jamming	Defensive; protects maneuver forces from RF-triggered weapons
Spot jamming	Jams one specific frequency (max power on target; ineffective vs. freq-hopping)
Sweep jamming	Covers a frequency range by sweeping; good vs. frequency-agile targets
Barrage jamming	Covers a large bandwidth at once; disperses power (power-hungry); good for many targets, bad for stealth
Follower jamming	Automatically targets receivers when a threat transmission is detected

- **Radar jamming: Noise jamming** (RF on victim's frequency to interfere with receiver gain) vs. **Deception jamming** (deceive radar discriminators/operator). Employment: **Support jamming** (SOJ/escort) and **Self-protection jamming.**

11.5 Countermeasures / Expendables

- **Chaff:** dispensed metallic strips that reflect EM waves; most widely used expendable EA; optimum length ≈ ½ **wavelength** of the victim radar frequency; first used WWII (RAF "WINDOW," Jul 1943). Uses: area saturation, chaff corridor.

- **Decoys:** replicate an electronic signature; saturate radar, coerce enemy to expose forces, defeat tracking.

11.6 EA Considerations

- De-confliction with friendly comms & EMS-dependent systems; may interrupt **intelligence collection** (CEMA controlling the EMS can take precedence over SIGINT collection); IRCs (MISO, MILDEC); non-hostile local EMS; hostile intelligence collection; **persistency** (jamming effects only last while the jammer emits & is in range).
- **Planning tools:** EWPMT, SPEED, GIANT.

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12. Electromagnetic Protection (EP) — (EA_ES_EP)

Source slide deck (PDF): EA_ES_EP.pdf

12.1 Definition

- **EP:** actions to protect personnel, facilities, and equipment from **friendly, neutral, enemy interference, and naturally occurring phenomena.**
- **EP vs. Defensive EA:** both protect friendly units in the EMS, but **EP acts on friendly systems**; **Defensive EA acts against enemy systems.**

12.2 EP Tasks (memorize the 6)

1. **Electromagnetic Hardening** (blanking, filtering, attenuating, grounding, bonding, shielding)
2. **Electromagnetic Masking** (controlled friendly radiation to protect emissions from enemy ES/SIGINT)
3. **Emission Control (EMCON)** (selective/controlled use of emitters to optimize C2 while minimizing detection/interference)
4. **Electromagnetic Spectrum Management**
5. **Wartime Reserve Modes (WARM)** (capabilities held in reserve, unused until wartime to preserve surprise)
6. **Electromagnetic Compatibility (EMC)** (systems operate in intended environment without unacceptable interference)

12.3 EP Techniques & TTPs

- Limiting transmissions, **Radio PACE plans**, terrain masking, directional antennas, spectrum management, frequency hopping, shielding, EMS management.
- **"The most basic protection from enemy EW is radio discipline"** — limiting transmissions, antenna masking, low power (Center for Army Lessons Learned).
- **Personal EP TTPs: Body Mass Shielding** (place body between jammer and receiver; rotate ~90°, wait ~2 min); **Terrain Masking** (use hills/buildings; place receiver near feature; put small GPS receivers in a **6–8 inch hole** to block EMI while keeping sky view).
- **Defeating jamming / improving Signal-to-Jamming Ratio:** increase transmitter power, adjust/relocate antenna, establish a **retransmission (RETRANS) station**, use an alternate route, **change frequencies.**

12.4 EMI Reporting

- **EMI:** any EM disturbance (intentional or unintentional) that degrades performance. Report via **Joint Spectrum Interference Report (JSIR)** through spectrum-manager channels — even if you can overcome it. Operators must be trained to identify jamming to report it.

12.5 Common Jamming Signals (recognition)

- **Random Noise** (like background noise); **Stepped Tones** (bagpipes); **Spark** (short high-intensity, rapid — very effective); **Gulls** (quick rise/slow fall — seagull cry); **Random Pulse**; **Wobbler** (howling); **Preamble Jamming** (mimics secure-net sync preamble).

12.6 EP Planning

- EP is as much **culture/discipline** as technical capability; a **team effort** (CEMA cell needs unit buy-in); **the single most important task of BDE-and-below CEMA — and the hardest.** Coordinate with **S-6** (network), **S-2** (threat), conduct EM vulnerability assessment, build/update **PACE plan**, incorporate jamming into all collective training.

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13. Electromagnetic Support (ES) — (EA_ES_EP)

Source slide deck (PDF): EA_ES_EP.pdf

13.1 Definition

- **ES:** actions tasked by / under direct control of an operational commander to **search for, intercept, identify, and locate/localize** sources of intentional & unintentional radiated EM energy — for **immediate** threat recognition, targeting, cueing EA, self-protection/EP, and supporting IRCs.

13.2 ES vs. SIGINT (HIGH-YIELD)

	ES	SIGINT
Legal authority	**Title 10**	**Title 50**
Purpose	Directly support operations (targeting, PIRs)	Build intelligence products
Data flow	Can pass info to SIGINT	Can pass "sterilized" info to ES

- ES and SIGINT are often co-located and mutually supporting; integrating EW + SIGINT is a **force multiplier** (reduces duplication, enables more collection). Task and purpose determine which asset is used.

13.3 ES Actions

- **Electromagnetic Reconnaissance** (detect/locate/identify/evaluate foreign EM radiations), **Electromagnetic Intelligence**, **Electromagnetics Security.**

13.4 ES Preparation

- **Electromagnetic Environment (EME) Survey** — like a weather report for the CEWO; begins with the enemy **Electronic Order of Battle (EOB).**
- **Enemy EOB:** comprehensive document of enemy electronic systems + associated platforms/frequencies; created/obtained by the **SIGINT TECH in the MICO**; one of the single most important documents for the EW PLT LDR.
- **Site Selection:** determine best location for ES based on terrain/elevation, enemy avenues of approach, signal coverage, comms support, concealment, logistics. Basic tools: **map, EOB, RF simulation software/coverage maps, enemy & friendly SITEMPs.**

13.5 Direction Finding (DF) — MEMORIZE

- **DF:** using RF equipment to locate signals of interest; provides **Line of Bearings (LOBs).**
- **LOB: single** DF sensor → approximate azimuth to the transmitter.
- **Cut: two** DF sensors → general location where two LOBs intersect.
- **Fix: three or more** DF sensors → location by triangulation.
- **Circular Error Probability (CEP):** the precise transmitter location cannot be certain (triangular area of overlap).
- **Path errors:** Scatter, Refraction, Reflection, Reradiating.
- Advanced DF methods: **TDOA (Time Difference of Arrival)**, **CI (Correlated Interferometry).**

BLOCK 6 — TACTICAL EMPLOYMENT OF THE EW PLATOON

14. EW Threat Assessment / IPOE (CYBOLC_EW_Threat_Class)

Source deck: *CYBOLC_EW_Threat_Class.pdf* — contains FOUO-marked slides — not hosted on the public site.

14.1 IPOE (Intelligence Preparation of the Operational Environment) — 4 Steps

1. **Define the battlefield environment.**
2. **Describe the battlefield's effects.**
3. **Evaluate the threat.**
4. **Determine threat COAs.**

- Analyze three layers: **Physical, Logical, Cyber-Persona** across Area, Structure, Capabilities, Organizations, People, Events.

14.2 Risk

- **Risk to Mission:** impacts the unit's ability to achieve objectives.
- **Risk to Force:** impacts personnel, equipment, sustainment.

- **Running Estimate:** current status snapshot updated at the end of every MDMP step; contains Facts, Assumptions, Specified/Implied tasks, Limitations/Constraints, RFIs. Answers: what the CDR needs to know from CEMA, what risks to mitigate, what info to track.

14.3 Example Threat

- **Aviaconversia EA system** — man/vehicle-portable Russian UHF jammer; jams 230 MHz & 433 MHz; **4 W**; range **150–200 km**; likely 3 (one per recon company).

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15. EW Platoon Force Structure & Mission (112-17ABOLC)

Source slide deck (PDF): Electromagnetic Warfare Platoon (PLT) Force Structure and Mission.pdf

15.1 Overview

- **EW PLT:** a **14–40 personnel** element of **MOS 17-series** soldiers, mounted or dismounted, supporting the unit's ability to **target (lethal/nonlethal), improve EMS situational awareness, and conduct ISR.**
- **Mission:** conduct **ES and EA** in close contact with enemy/civilian populations to enable maneuver commanders to seize, retain, and exploit the initiative.
- **Location:** in **HHC BDE** (ABCT/SBCT) or the **Multi-Functional Reconnaissance Company (MFRC)** (MBCT). Falls under Brigade HQ, reporting to the BCT Commander.

15.2 Command/Support Relationships

- **Organic** to HHC BDE / MFRC. **ADCON/garrison** — Brigade HQ. **OPCON** — Brigade S-3 (or designated OPCON unit). **Technical control** — Brigade **CEMA cell**. Supports/coordinates with S-2 and Fires.

15.3 Key Personnel

- **Platoon Leader (17B):** responsible for all the platoon does/fails to do — training, employment, personnel, logistics. **Platoon Sergeant:** second-in-charge, senior enlisted. **Team Chief:** mission accomplishment, training, maintenance, tactical employment, calling/adjusting indirect fires. **Team Members:** proficiency in MOS tasks; detect/classify/report threat emissions.

15.4 The "So What" of EW PLTs

- **Can do:** sense signals, geolocate signals, conduct EMS baseline, conduct localized EA (jamming), provide SA in the close area.
- **Can't do:** break radio encryption*, sense/jam an entire BDE AO, sense at infinite distance.
- **Should do:** be placed **near or beyond the FLOT**, operate with recon elements, provide actionable data (NAI/PIR), **tip and cue** other sensors.
- **Shouldn't do:** be the primary collection system, get an entire AO, jam outside emergency/decisive situations, be placed at the rear/near CPs, target radars, be a long-term collection asset.
- **Needs to succeed:** be semi-autonomous, provide own local security, have a validated **PACE plan (incl. BLOS data)**, move/establish sites tactically, understand targeting/recon.

- **Needs from the unit:** extended logistics, area security, an assigned NAI/local AO, refined PIRs, target attack guidance matrix, EOB with specific SOIs, clear CDR's intent.

15.5 EW Formations by Echelon

- **Division EW Company** (organic to DIV I&EW BN) — close-area support; includes a **Defensive EA PLT** (protects vs. artillery fuses, RCIEDs, spectrum-homing rockets/missiles, observation & EO devices).
- **Corps EW Company** (organic to Corps MI BDE) — long-range sensors, **deep fight**; includes an **Operations PLT** (planning/management).
- **SOF EW PLTs** — each SF group has an organic EW PLT; teams deploy independently.
- **MDTF EW PLT** — extended-range sensing/effects; two EW sections + aerial multi-functional EW section; LOS/BLOS EA.

15.6 EW Core Competencies & Fundamental Principles (ATP 3-12.4)

- **Core Competencies:** (1) **Drive understanding**, (2) **Protect friendly personnel & capabilities**, (3) **Deliver effects.**
- **Fundamental Principles: Operational focus**, **Adaptability & versatility**, **Global reach.**
- **Nesting:** Execute Missions (EA/ES/EP) → Exercise Fundamental Principles → Focus on Core Competencies → Achieve desired End State (controlled use of the EMS).

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16. EW PLT Movement Techniques (112-17ABOLC)

Source slide deck (PDF): Electromagnetic Warfare PLT Movement Techniques.pdf

16.1 Combat Power & Mobility

- **EW PLT is NOT designed to close with and destroy the enemy** — it prioritizes **security and stealth**, avoiding/engaging the enemy only when necessary.
- Doctrine: **3:1 advantage** generally required to maneuver against and destroy an enemy force.
- MFRC EW PLT can maneuver against ~a **10-man** enemy unit; an EW **Team** vs. a 1–2 man unit. Rifle PLT can take a 12-man unit.
- **M249** provides more suppression than M4 but limited ability vs. enemy in cover; largest EW PLT vehicle caliber is **5.56 mm** (only 3 of 4 vehicles have mountable weapons).
- **M2A3 Bradley (BFV):** 3 crew + 6 dismounts; 25mm M242 Bushmaster, 7.62mm M240C, 2× TOW; 500-hp engine, 34–37 mph.

16.2 Operations Methodology

- **Centralized:** PLT operates as a whole under the EW PL.
- **Decentralized:** PLT parceled out in team/squad elements as **TACON** to maneuver elements (PL retains ADCON/OPCON).
- **Base dismounted element = the EW team** (train it most).

16.3 Formations & Movement

- **Echelons:** PLT (14–30, LT), Squad (9, SSG), Team (4, SGT).
- **3 most common formations: Column/File** (speed + control, less firepower), **Line** (firepower forward, less speed), **Wedge** (360° security/flexibility, default for movement).
- **Mounted:** Staggered Column (speed/limited lateral dispersion), Line (assault/open areas — poor for DF-on-the-move), Wedge (open terrain, vague enemy situation).
- **Movement techniques (3): Traveling** (speed, contact unlikely), **Traveling Overwatch** (contact possible), **Bounding Overwatch** (contact expected).

16.4 Actions on Contact (4 steps)

1. **React** (battle drill; report to higher). 2. **Develop the situation.** 3. **Choose an action** (Attack, Bypass, Defend, Delay, Withdraw). 4. **Execute and report.** Maintain contact unless told to break contact.

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17. EW Site Selection (112-17ABOLC)

Source slide deck (PDF): Electromagnetic Warfare Site Selection.pdf

17.1 EW Site Requirements (memorize the 5)

1. **Line of Sight (LOS) to Objective** — need EMS observation (detect/jam SOIs), not necessarily visual.
2. **Accessible** — can you and support assets reach it.
3. **Tenable** — can you sustain/support it and retain the ability to fall back/break contact. *(Never place a team where it can be easily isolated.)*
4. **Secure** — ability to provide security (avg EW team = M4s + M249).
5. **Supports Communications** — timely, accurate reporting (PACE, BLOS).

17.2 Requirements vs. Considerations

- **Requirements** = the selection checklist; **Considerations** = factors that let you evaluate the requirements (mission-dependent).
- **Considerations:** Terrain, Mobility, Sustainment, Duration, External Support, Equipment Capabilities, Enemy.
- **Terrain analysis acronym: OACOK** — Obstacles, Avenues of Approach, Cover/Concealment, Observations/fields of fire, Key terrain.
- **Day of Supply (DOS):** standard measure of average daily supply expenditure.

17.3 Key Inputs

- **Enemy EOB** (higher-level context — the "enemy SITEMP in the EMS") and the **SOI list** (specific target frequencies) are the **two most important documents** for an EW mission.
- Tools: **Maps** (map recon) + **Simulation devices** (every EW PLT has one; requires power) → analog product.

17.4 EMS Survey

- A broad observation of the EMS at a location using a **spectrum analyzer** to "sweep the EMS." Detected signals checked against operational products (friend/neutral/foe). **An EW site is not established until an EMS survey is conducted.**
- **Survey steps (EWTS):** move to good "view" of the EME → note terrain/location/weather → set analyzer to max viewable/assigned bandwidth → set **max-hold** to catch intermittent signals → capture/characterize bands → narrow to ~20–50 MHz → step through capturing each emission (center freq, bandwidth, S/N, receive power, shape, behavior) → log everything → save final report.

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18. Establishing an EW Hide Site / Observation Post (112-17ABOLC)

Source slide deck (PDF): Establishing an Electromagnetic Warfare Hide Site.pdf

18.1 Observation Post (OP)

- **OP:** a position from which observations are made or fires are directed/adjusted.
- Two selection questions: best location to get the information? best way to get it **without being compromised**?
- **Do NOT skyline observers** — avoid hilltops; use the slope/side. Provide covered/concealed routes, overlapping fields of observation.
- **Occupation order:** leader's recon → occupy → **secure site (weapons emplaced, 360° security)** → **conduct EMS survey** → OP established.
- **Manning:** minimum **2 soldiers**; ≥12 hours requires team/squad + rotation plan.

18.2 Camouflage, Cover & Concealment

- **Visual camouflage** priority — visual observation is the quickest way an OP becomes decisively engaged.
- **Hide Site:** a concealed area for equipment/sleep/eat areas not immediately needed at the OP (more equipment at the OP = bigger target).
- **EMS Camouflage:** EM waves interact 6 ways (Transmission, Reflection, Refraction, Diffraction, Absorption, Scattering).
- **Knife's Edge Diffraction:** signal bends around an obstacle → you can be physically covered and still observe a signal (no visual LOS needed).
- **Terrain Masking:** place comms equipment behind terrain so only friendly forces can see the signal.
- **Directional Antennas:** decrease interception by the enemy DF team.
- **Remoting Antennas:** separate the antenna from the radio (wired/wireless) → keep soldiers/equipment behind cover while transmitting.
- **Thermal Signatures:** heat is visible in the IR band — manage thermal signature.
- **EW site establishment sequence (CEWTS):** receive mission/TLPs → confirm PACE → enforce EP → maneuver & establish site → conduct EW IAW **AGM (Attack Guidance Matrix)** → determine re-attack → assess EP → recover equipment & sanitize site → provide **EWMSNSUM** report.

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19. Fundamentals of EM Reconnaissance (112-17ABOLC)

Source slide deck (PDF): Fundamentals of Electromagnetic (EM) Reconnaissance.pdf

19.1 Definitions

- **Reconnaissance:** a mission to obtain information about enemy/area by visual observation or other detection methods.
- **Electromagnetic reconnaissance:** detection, location, identification, and evaluation of foreign EM radiations — one of the most common EW PLT missions.
- Integrating processes: **IPOE, Targeting, Information Collection.**

19.2 Techniques, Types, Methods

- **Techniques: Reconnaissance-push** (thorough understanding → push assets to confirm/deny assumptions) vs. **Reconnaissance-pull** (uncertain situation → recon develops the picture and pulls the main body toward least resistance).
- **Types:** Zone, Area, Route, Reconnaissance in Force, Special Reconnaissance.
- **Methods:** Mounted, Dismounted (most detailed, most time-consuming), Aerial, Reconnaissance by Fire.

19.3 Recon vs. Surveillance

- **Surveillance:** systematic **passive, continuous** observation. **Reconnaissance: active**, usually includes human participation and fighting for information. They complement/cue each other.

19.4 The 7 Fundamentals of Reconnaissance (MEMORIZE)

1. Ensure continuous reconnaissance.
2. Do not keep reconnaissance assets in reserve.
3. Orient on the reconnaissance objective.
4. Report all required information rapidly and accurately.
5. Retain freedom of maneuver.
6. Gain and maintain enemy contact.
7. Develop the situation rapidly.

- **LTIOV (Latest Time Information is Of Value):** deadline tied to a decision point.

19.5 EMS Reporting

- No universal format — units develop **SOPs.** Report content: **Emitter Data** (carrier freq, bandwidth, signal characteristics) and **Geolocation Data** (estimated location, Fix/Cut/LOB, CEP).
- **Report types:** Stop Jamming Message, **EW Frequency Deconfliction Message** (prevents frequency fratricide), **EW Mission Summary (EWMSNSUM)**, **EW Tasking Message**, **JSIR** (Joint Spectrum Interference Resolution), **MIJIFEEDER** (Meaconing, Intrusion, Jamming, Interference — immediate tactical EMI report).
- **Non-standard quick reports: FAP (Frequency-Azimuth-Power)** and **EW GEO Report.**
- **Sensor-to-effect chain:** sensor picks up signal → send **Emitter report** to SIGINT/CEMA → geolocate → **tip/cue** other ISR → match to SOI/EOB → **SPOT report / fire mission** → answer PIRs → enable CDR

decisions.

20. EW Training Management (112-17ABOLC)

Source slide deck (PDF): Electromagnetic Warfare Training Management.pdf

20.1 METL

- **METL (Mission-Essential Task List):** tasks an organization must train to be proficient in its mission. Determined by **Department of the Army** (with proponent input). Found at the **Army Training Network (ATN).**
- **Reference sources: FMSWeb** (MTOE — manning/equipment), **ATN** (METL & CATS), **CATS** (Combined Arms Training Strategy) → contains the **T&EO (Training and Evaluation Outline).**
- If no METL: PL/PSG conduct a **METL crosswalk** (FM 7-0), aligning to the higher echelon's plan.

20.2 Training Strategies

- **CEWTS (TC 3-12.1.2.2):** Cyber & EW Training Strategy — training standards; supports the 2018 NDS ("gain/maintain information advantage in cyberspace, space, and the EMS").
- **EWTS (FORSCOM):** EW training/certification for units with Quick Reaction Capabilities (QRC); adaptive for near-peer/LSCO.
- **EW Program of Record equipment: EWPMT v1**; most PLTs have COTS/GOTS (e.g., VROD/BEAST).

20.3 8-Step Training Model

1. Plan the training event → 2. Train and certify leaders → 3. Recon training sites → 4. Issue the OPORD → 5. Rehearse → 6. Train → 7. Conduct After Action Reviews → 8. Retrain.

20.4 Troop Leading Procedures (TLPs) — 8 Steps

1. Receive the mission → 2. Issue a WARNORD → 3. Make a tentative plan → 4. Initiate movement → 5. Conduct reconnaissance → 6. Complete the plan → 7. Issue the order → 8. Supervise and refine.

- **METT-TC (I):** Mission, Enemy, Terrain, Troops & support available, Time available, Civil considerations, (Informational considerations). Used during mission analysis in Step 3.
- Time management: **one-third / two-thirds rule** (leader uses 1/3 of available time for planning, gives 2/3 to subordinates).

21. Operating in a Contested Space Environment (CCoE)

Source slide deck (PDF): CCoE Operating in a Contested Space Environment (2).pdf

21.1 Space Fundamentals

- The Army is the **largest user of space-enabled capabilities**, integrated into all six WfFs. A BCT has **>3,200 GPS-enabled** and **>300 SATCOM-enabled** pieces of equipment.
- **D3SOE = Denied, Degraded, and Disrupted Space Operational Environment** — a **condition** of the operational environment.
- **PNT (Positioning, Navigation, Timing):** each GPS receiver needs LOS to a minimum of **4 satellites** (8–11 usually in view) for latitude, longitude, altitude, and receiver clock/time. GPS is an easy target — very weak signal (~50-watt light bulb), known frequencies, cheap jammers. Primary threat = **ground-based jammers targeting receivers.**
- GNSS constellations: GPS, GALILEO, GLONASS, BeiDou, NavIC, QZSS.

21.2 Threats & Indicators

- **Primary concern for Army tactical units:** SATCOM jammers, GPS jammers, communications interference, EMP.
- **GPS interference indicators:** DAGR "Jamming Detected" banner, loss of 4+ satellites, stale FFT icon, gumball/stoplight turning red/amber.
- **SATCOM indicators:** static, bleed-over, loss of comms. **PGM indicators:** rounds miss or become duds.

21.3 Mitigation & Takeaways

- **EMCON, redundancy/layered capabilities, spectrum management/dynamic spectrum access, hardening, deception/signature management, alternate nav/timing, EW, training for degraded conditions, ISR/threat awareness.**
- **Takeaway framework: PREPARE > RECOGNIZE > REACT > REPORT.** No civilian GPS receivers; **Encrypt!**; know your equipment (contested vs. uncontested); do terrain analysis; have a PACE plan; body-shield/dig a hole; report EMI to S2/S6/EW.
- **Ukraine lessons:** minimize emissions; reduce PNT dependency; reintroduce ground EW (**Terrestrial Layer System — TLS** primary EW/SIGINT weapon system); intensive drone/UAS use; **Starlink** resilience; geolocation risk from cell phones/social media (Russian **Leer-3** picks up 2,000+ phones within 3.7 mi).

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22. Case Study — 2nd Nagorno-Karabakh War 2020 (CALL 21-655)

Source slide deck (PDF): 21-655-nagorno-karabakh-2020-conflict-catalog-aug-21-public.pdf_safe.pdf

- **27 Sep–10 Nov 2020:** Azerbaijan vs. Armenia over Nagorno-Karabakh (Artsakh). Decisive **Azerbaijani victory**; Russian-brokered cease-fire (2,000 peacekeepers).
- Azerbaijan overpowered Armenia with advanced **UAS** (Turkish **Bayraktar TB2**, Israeli **Harop/SkyStriker** loitering munitions) + SOF to identify targets for artillery/missiles/loitering munitions.
- Tactic: Soviet-era **AN-2 biplane decoys** forced Armenian air defenses to reveal themselves; TB2s above targeted radars/launchers. Armenian Soviet-era air defense couldn't detect TB2s at altitude. Armenia used Russian **Polye-21 EW** (disrupted UAS ~4 days but was targeted).
- **Information campaign:** full-motion video of strikes released on social media to demoralize.

- **Lessons for ULO:** expect UAS proliferation (120+ nations have military UAS); air defense must detect/defeat wide variety of UAS; **master the fundamentals** (camouflage, masking, dispersion, jam/disrupt sensors, multispectral decoys); aggressive information operations.

QUICK-REFERENCE TABLES

A. The Three Divisions of EW at a Glance

Division	Definition	Key Sub-Tasks
EA (Attack)	EM/directed energy/anti-radiation weapons to degrade/neutralize/destroy enemy (a form of fires)	Jamming, deception, intrusion, probing, EMP, countermeasures (chaff/decoys)
EP (Protection)	Protect friendly from friendly/neutral/enemy/natural interference	Hardening, Masking, EMCON, Spectrum Mgmt, WARM, EMC
ES (Support)	Search/intercept/identify/locate radiated EM energy for immediate use	EM Reconnaissance, EM Intelligence, EM Security, DF (LOB/Cut/Fix)

B. Direction Finding Sensors

# of DF Sensors	Result	Definition
1	**LOB**	Approximate azimuth to transmitter
2	**Cut**	General location (LOB intersection)
3+	**Fix**	Precise location via triangulation

C. Must-Memorize Formulas

Quantity	Formula
Speed of light	$c = 3 \times 10^8$ m/s
Velocity	$V = \lambda F$
Frequency	$F = c/\lambda$; $F = 1/T$
Wavelength	$\lambda = c/F$
Decibels	dB = 10 log(P_out/P_in)
dBm	10 log(P in mW)
dBW	10 log(P in W)
dBW↔dBm	dBW = dBm − 30
dB → ratio	ratio = 10^(N/10)
Loss	α(dB) = −g(dB)
Resonant length	L = C/(2f)
Nyquist	sample rate ≥ 2 × f_max

D. dB Shortcuts

Change to ratio	Change in dB
× 2	+ 3 dB
÷ 2	− 3 dB
× 10	+ 10 dB
÷ 10	− 10 dB

E. Key Acronyms

CEMA (Cyberspace Electromagnetic Activities) · EW (Electromagnetic Warfare) · EA/EP/ES · EMS (Electromagnetic Spectrum) · EMSO (EM Spectrum Operations) · SMO (Spectrum Mgmt Operations) · EMCON (Emission Control) · WARM (Wartime Reserve Modes) · EMC (EM Compatibility) · EMI (EM Interference) · SIGINT · DF · LOB/Cut/Fix · CEP (Circular Error Probability) · EOB (Electronic Order of Battle) · SOI (Signal of Interest) ·

JRFL (Joint Restricted Frequency List) · JSIR · MIJIFEEDER · EWMSNSUM · IPOE · METT-TC(I) · OACOK · METL/CATS/T&EO · TLP · PACE · FLOT · NAI/TAI · PIR/CCIR · LTIOV · EWPMT/SPEED/GIANT · TLS · D3SOE · PNT/GNSS · SEAD · RCIED/CREW · SDR · IQ · S/N.

SELF-TEST QUESTION BANK

(Cover the answers; test yourself. Answers follow each question.)

History & Doctrine 1. What famous incident saw the first deliberate radio jamming? → **1901 America's Cup Yacht Race.** 2. What conflict saw the first combat use of deliberate jamming? → **Russo-Japanese War (1904).** 3. What does RADAR stand for? → **RAdio Detection And Ranging.** 4. What was DoD's response to the RCIED threat? → **The CREW program (Counter-RCIED Electronic Warfare).** 5. When did EW officially become part of the Cyber Branch? → **1 October 2018.** 6. Name the three divisions of EW. → **Electromagnetic Attack, Electromagnetic Protection, Electromagnetic Support.** 7. What is CEMA comprised of? → **EW + Cyberspace Operations + Spectrum Management Operations.** 8. Which manual is the Army's core CEMA/EW publication? → **FM 3-12.**

EMS & Waves 9. Frequency and wavelength are related how? → **Inversely proportional** (high freq = short λ). 10. List the RF sub-bands low→high. → **ELF, SLF, ULF, VLF, LF, MF, HF, VHF, UHF, SHF, EHF.** 11. What frequency band is GPS in? → **UHF.** 12. Speed of light? → **3×10^8 m/s.** 13. Name the 4 core EM wave interactions. → **Reflection, Refraction, Diffraction, Absorption.** 14. Doppler: moving toward the receiver produces what shift? → **Blue shift (higher frequency).** 15. How many ionosphere layers day vs. night? → **4 day, 2 night.** 16. An EM wave requires a medium — true or false? → **False** (self-propagating).

EW Math 17. dB formula? → **10 log(P_out/P_in).** 18. Doubling the power ratio changes dB by? → **+3 dB.** 19. Convert 30 dBW to dBm. → **60 dBm** (dBm = dBW + 30). 20. A device has P_in 10 W, P_out 1 W — loss in dB? → **10 dB.**

Systems 21. Full-duplex requires how many frequencies? → **Two.** 22. Most common tone-squelch technology? → **CTCSS.** 23. Jamming affects transmitters or receivers? → **Receivers.** 24. Resonance occurs at what fraction of the target wavelength? → **½ wavelength.** 25. Cross-polarization (vertical vs. horizontal antenna) causes what loss? → **~30 dB.** 26. HackRF One frequency range? → **1 MHz – 6 GHz.** 27. Nyquist sampling rate? → **≥ 2× the highest frequency.** 28. Which antenna is the theoretical omnidirectional benchmark? → **Isotropic (0 dBi).**

EA/EP/ES 29. For a receiver to detect a signal, S/N must be? → **Greater than 1.** 30. Spot vs. barrage jamming? → **Spot = one specific frequency; Barrage = wide bandwidth, dispersed power.** 31. Optimum chaff length? → **~½ wavelength of the victim radar frequency.** 32. The three types of electromagnetic deception? → **Simulative, Manipulative, Imitative.** 33. Name the six EP tasks. → **Hardening, Masking, EMCON, Spectrum Management, WARM, EMC.** 34. "The most basic protection from enemy EW is…" → **Radio discipline.** 35. ES falls under which Title; SIGINT under which? → **ES = Title 10; SIGINT = Title 50.** 36. What report is used for immediate tactical EMI? → **MIJIFEEDER** (formal resolution via **JSIR**).

Tactical Employment 37. The 4 steps of IPOE? → **Define environment, Describe effects, Evaluate threat, Determine threat COAs.** 38. The 5 EW site requirements? → **LOS to objective, Accessible, Tenable, Secure, Supports Communications.** 39. Terrain analysis acronym? → **OACOK.** 40. When is an EW site officially established? → **After an EMS survey is conducted.** 41. The two most important documents for an EW mission? → **Enemy EOB and the SOI list.** 42. The 7 fundamentals of reconnaissance (first three)? → **Ensure continuous**

recon; do not keep recon in reserve; orient on the objective. 43. Surveillance vs. reconnaissance? → **Surveillance is passive/continuous; reconnaissance is active and fights for information.** 44. The 8 steps of TLPs (first three)? → **Receive the mission, Issue a WARNORD, Make a tentative plan.** 45. METT-TC(I) stands for? → **Mission, Enemy, Terrain, Troops/support, Time, Civil considerations, (Informational).** 46. Where is the EW PLT located in an MBCT? → **The Multi-Functional Reconnaissance Company (MFRC).** 47. Default dismounted movement formation? → **Wedge** (360° security). 48. Three movement techniques by contact likelihood? → **Traveling, Traveling Overwatch, Bounding Overwatch.** 49. D3SOE stands for? → **Denied, Degraded, and Disrupted Space Operational Environment.** 50. Minimum satellites a GPS receiver needs? → **Four.** 51. Contested-space takeaway framework? → **Prepare > Recognize > React > Report.** 52. Who won the 2020 Nagorno-Karabakh War and with what decisive capability? → **Azerbaijan, using UAS/loitering munitions (Bayraktar TB2, Harop) integrated with SOF and fires.**

End of Study Guide. Study blocks 3 (EW Math) and 5 (EA/EP/ES) are historically the heaviest exam material — prioritize the formulas, the dB shortcuts, the jamming techniques, and the LOB/Cut/Fix definitions.